LUNAR MODULE

STRUCTURES, MECHANICAL SYSTEMS AND ELECTROEXPLOSIVE DEVICES

COURSE NO. 30915

STUDY GUIDE

FOR TRAINING PURPOSES ONLY

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During earth prelaunch activities, various components and areas must be readily available; therefore, adequate access panels, in the outer skin and insulation, are readily available within these areas for easy access.

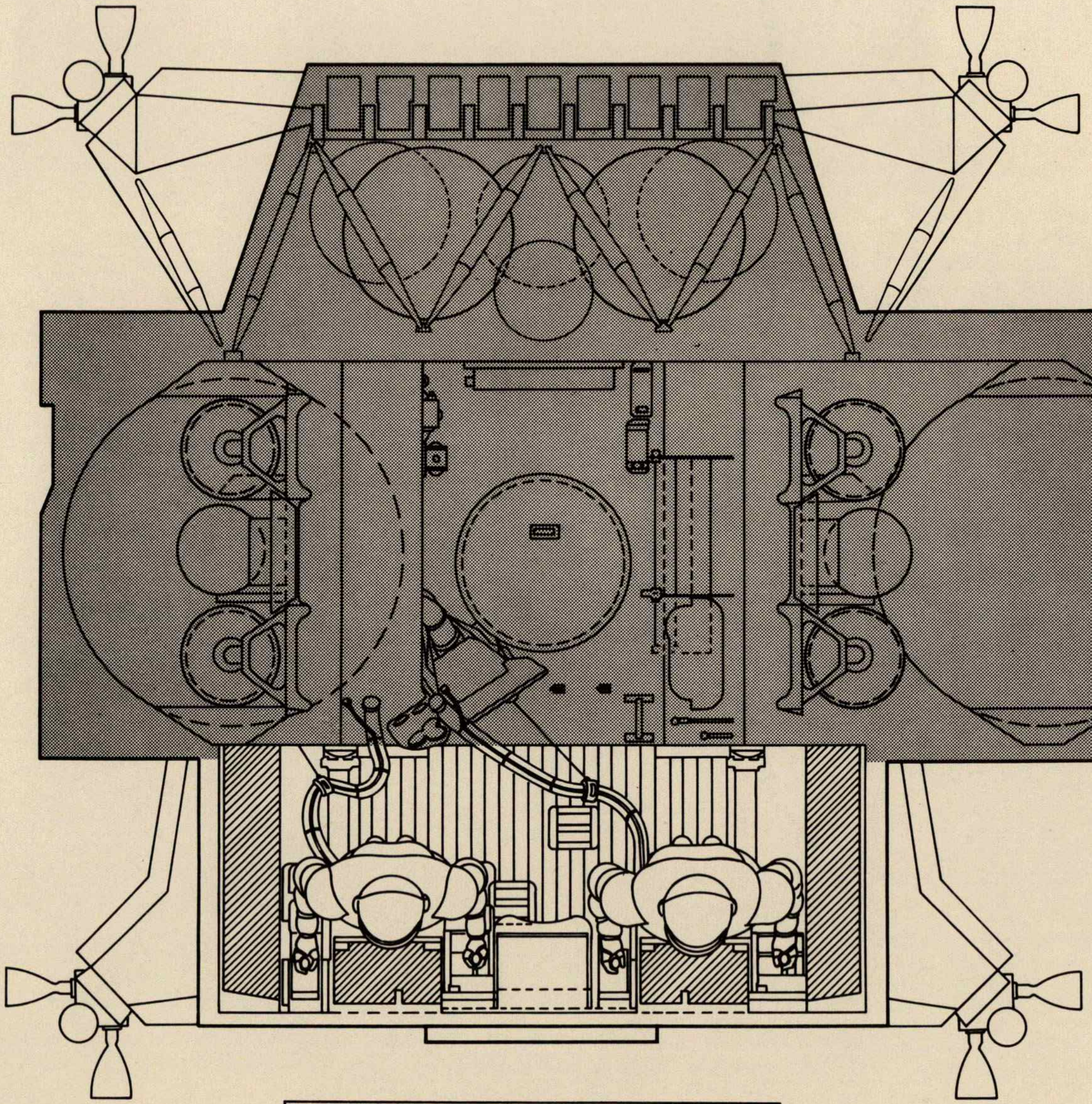
CREW COMPARTMENT

Forward Cabin Area: As previously stated, the crew compartment is the frontal area of the Ascent Stage, being approximately 92 inches in diameter and 42 inches in depth, creating an adequate work area for the two astronauts. (Figures 11 and 12.) This area having the flight stations with the instrument panels, armrests and body restraints, landing aids, two front and docking windows, alignment optical telescope and switch/circuit breaker panels. Flight stations are 44 inches apart; each astronaut has a set of controllers and armrests (Figure 13). The armrests have adjustments to accommodate all required percentile anthropometric sizes when the proper shoes are worn. A design reference point (design eye) orients flight station geometry for the required balance between external visibility, visual and physical access to controls and displays. Instrument panels, windows, glare shields, controllers, armrests and landing aids are oriented with respect to this design reference point. The flight station is provided with the necessary floor to overhead height, body clearance and armrest adjustment to accept all astronauts in the 10 to 90 percentile anthropomorphic body size. An astronaut need not be indexed at the design eye in order to effectively perform his assigned tasks; this requires that no adjustment with respect to the floor, is operationally employed for this purpose. The docking design eye orients the overhead docking window with respect to the Commander's flight station viewing angle. The main instrument panels are cantered and centered between the two flight stations to permit sharing and easy scanning by the astronauts. (Figures 14 and 15)

The optical alignment station is in the center aisle between the individual flight stations, for use of the alignment optical telescope (OAT). A single step-up adjustment is used in order to accommodate the range of percentile astronauts to the eyepiece and AOT controls (Figure 16).

The PLSS donning station (Figure 17) is also in the center aisle but slightly aft of the optical alignment station. A removable, height adjusting PLSS supporting harness, permits donning and doffing of the PLSS backpack. There is a recharge station on the -Y side of the equipment tunnel for recharging the PLSS water and oxygen, (Figure 18) completely accessible from the forward cabin. Tunnel design and equipment permits space for resting or sleeping (Figure 19).

Figure 11



ASCENT STAGE PLAN VIEW
(CREW COMPARTMENT)

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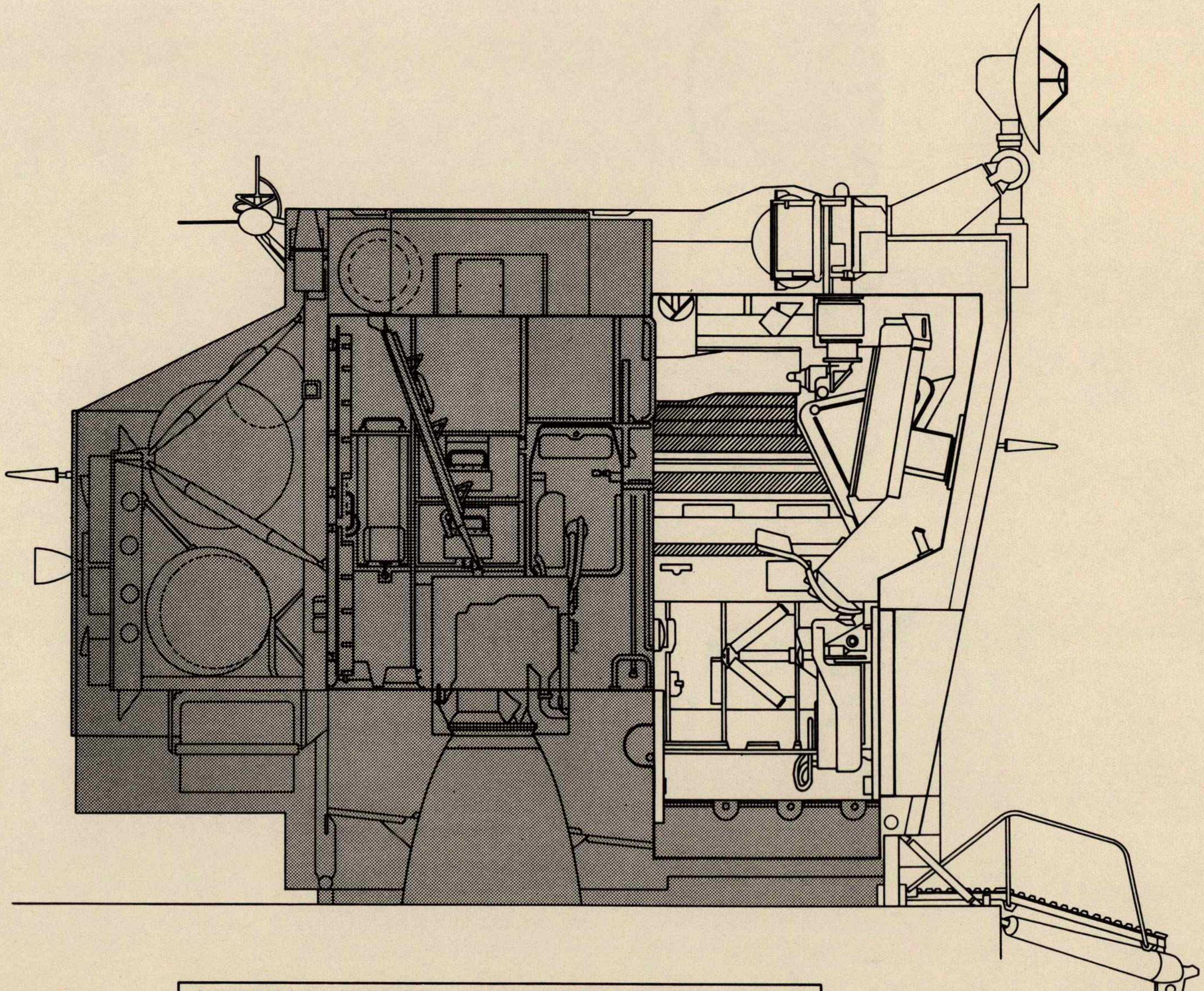
Figure 12



ASCENT STAGE CUTAWAY-LEFT SIDE
(CREW COMPARTMENT)

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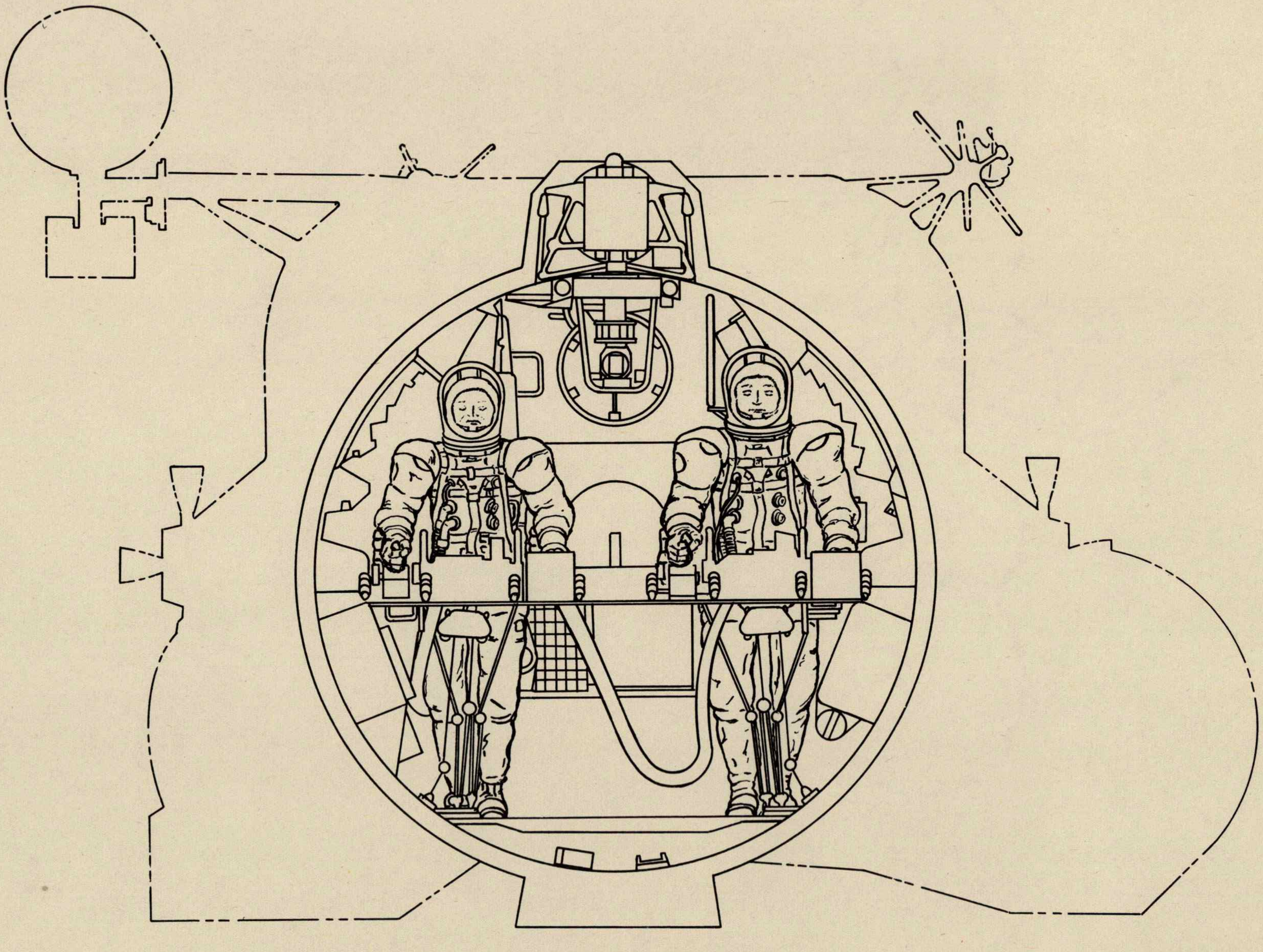

Figure 13

FLIGHT STATION (LOOKING AFT)

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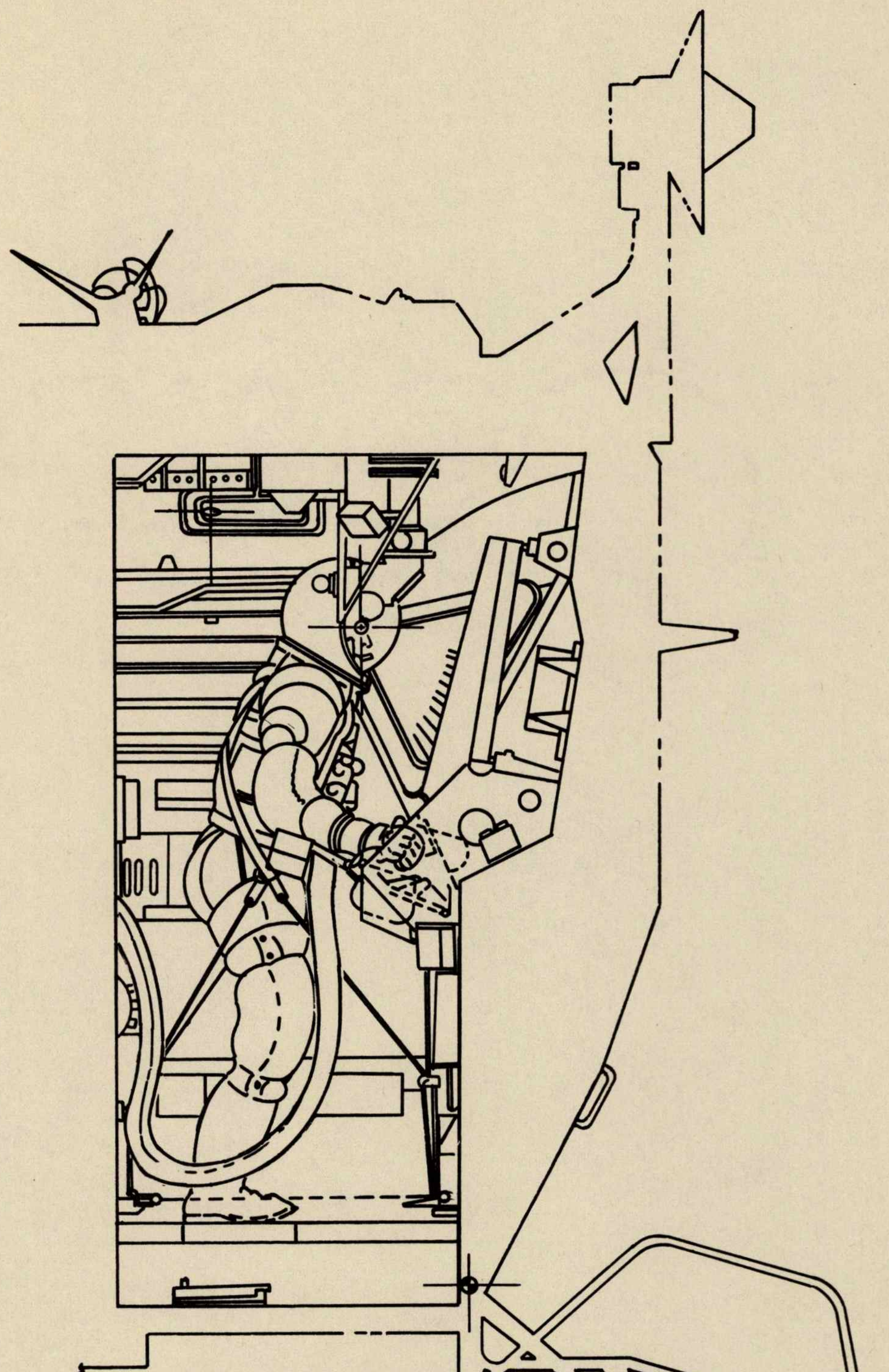

ASTRONAUT IN FLIGHT POSITION

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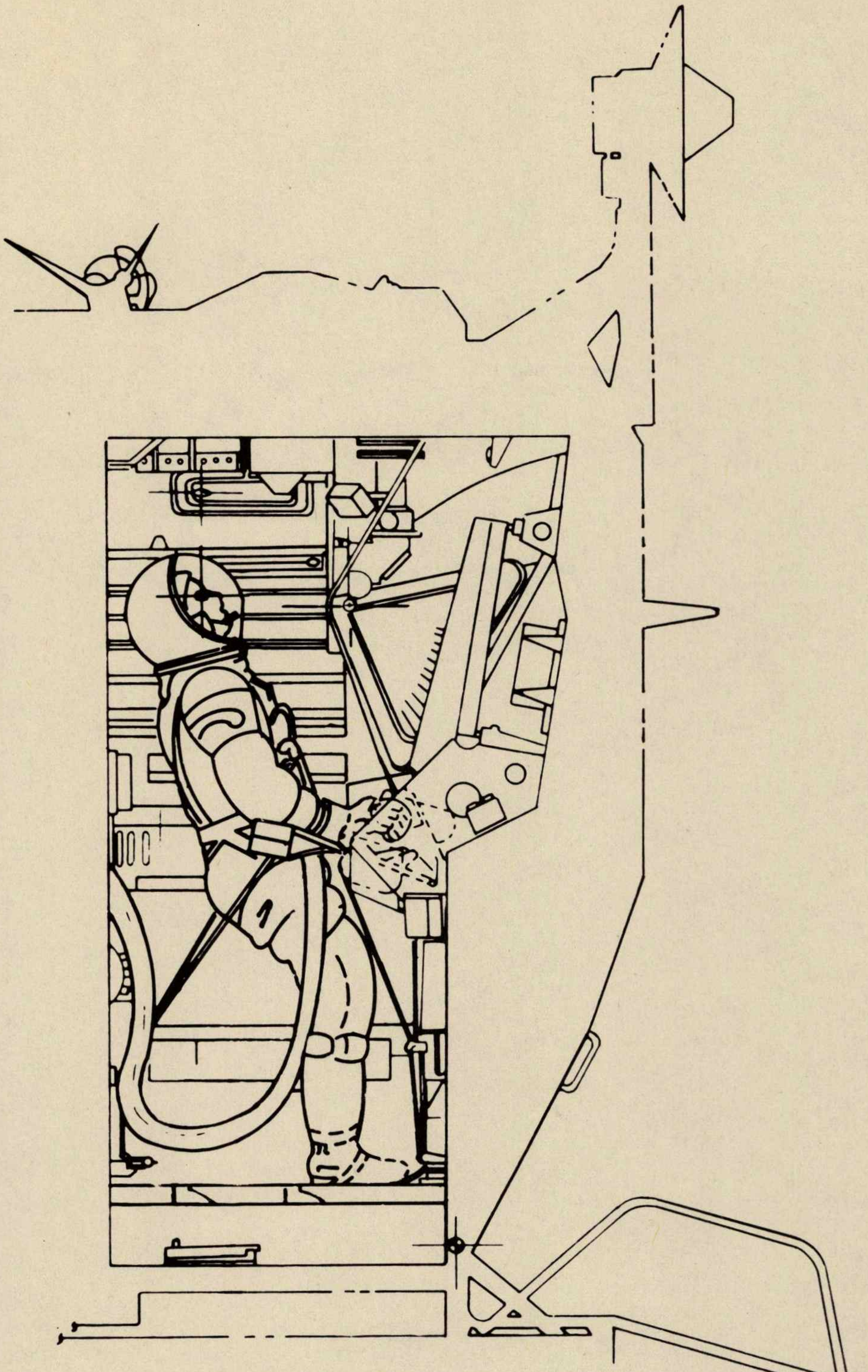
Figure 14



ASTRONAUT IN DOCKING WINDOW POSITION

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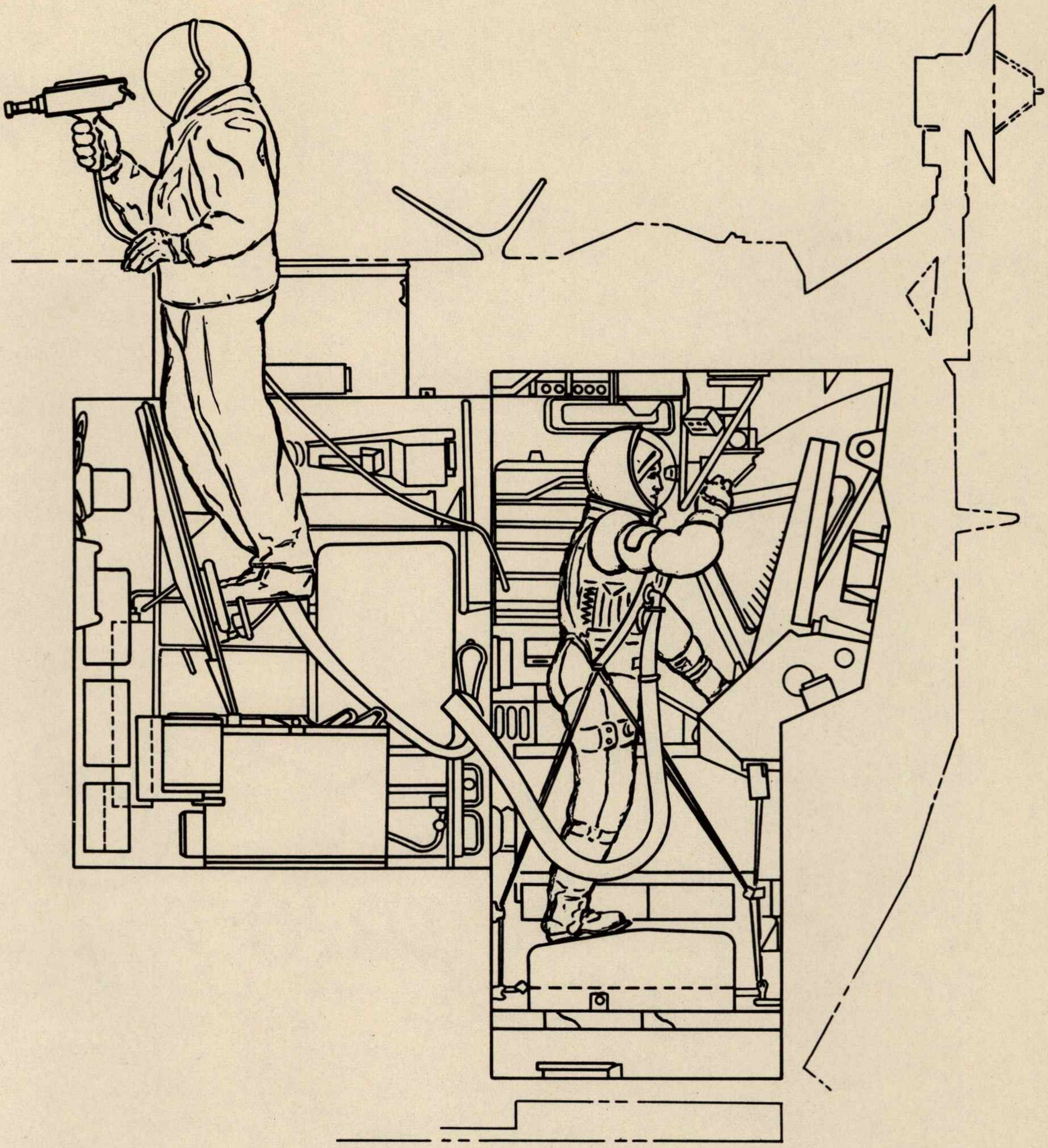
Figure 15



ASTRONAUTS USING AOT AND TV CAMERA

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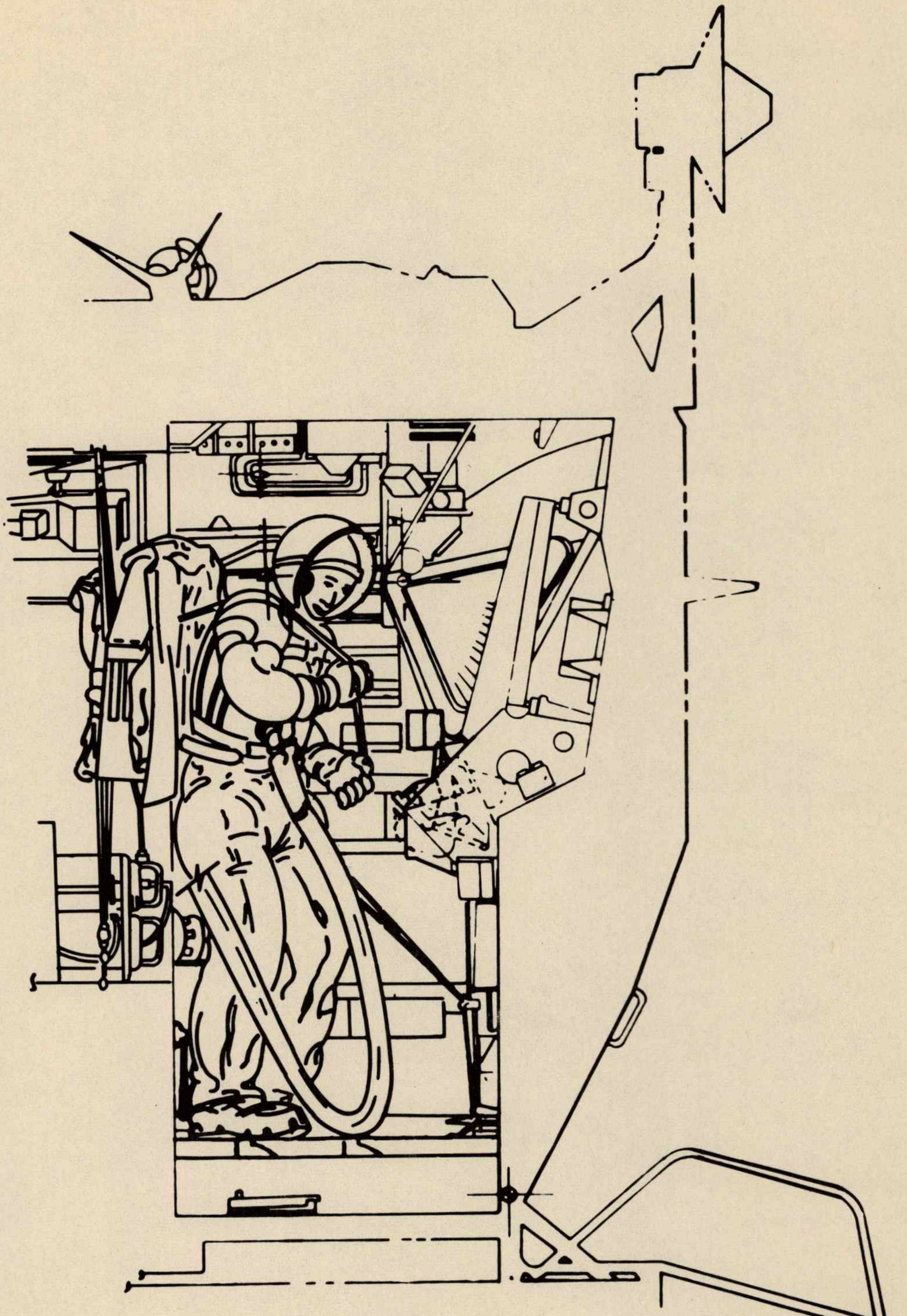

DONNING STATION

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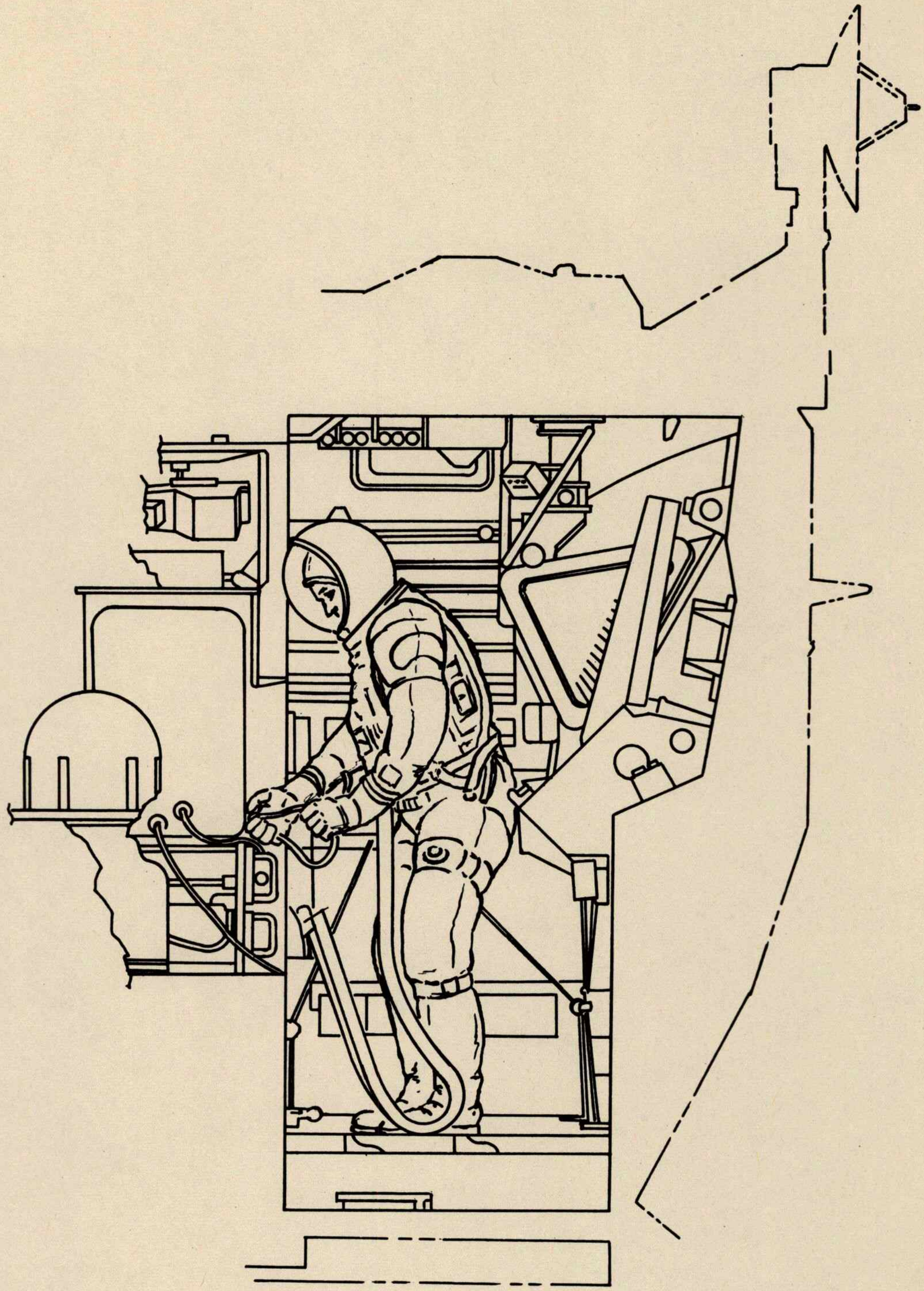
Figure 17



RECHARGE STATION

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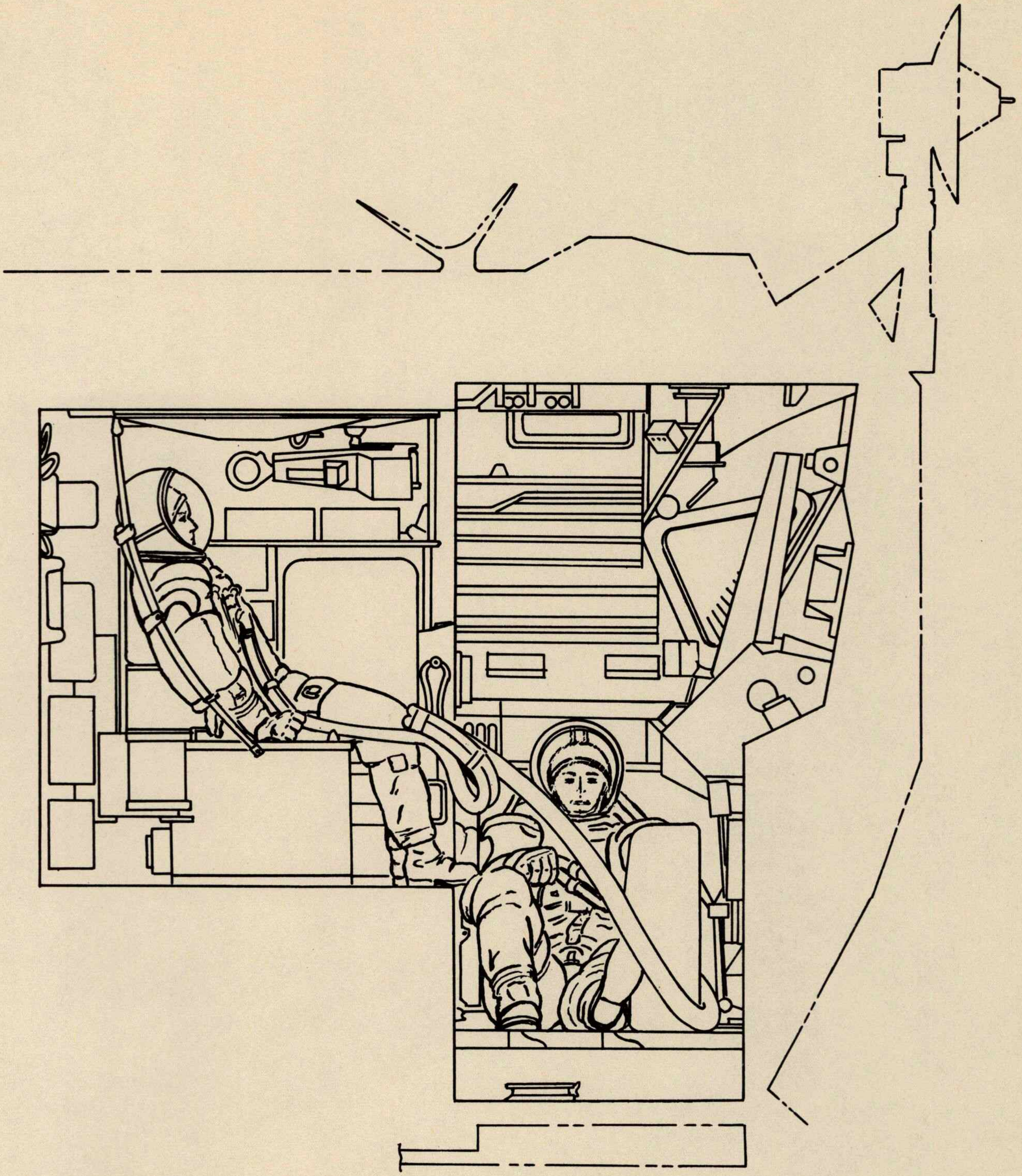
Figure 18



ASTRONAUTS-REST POSITIONS

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Control and Display Panels: The control and display area consists of twelve consoles and panels: (Figure 20) two main display panels (1 & 2), cantered forward 10°; two lower center panels (3 & 4), sloping down and aft 45° towards the horizontal; two bottom side panels (5 & 6); two lower side consoles (8 & 12); two center side consoles (10 & 14), and two upper side consoles (11 & 16).

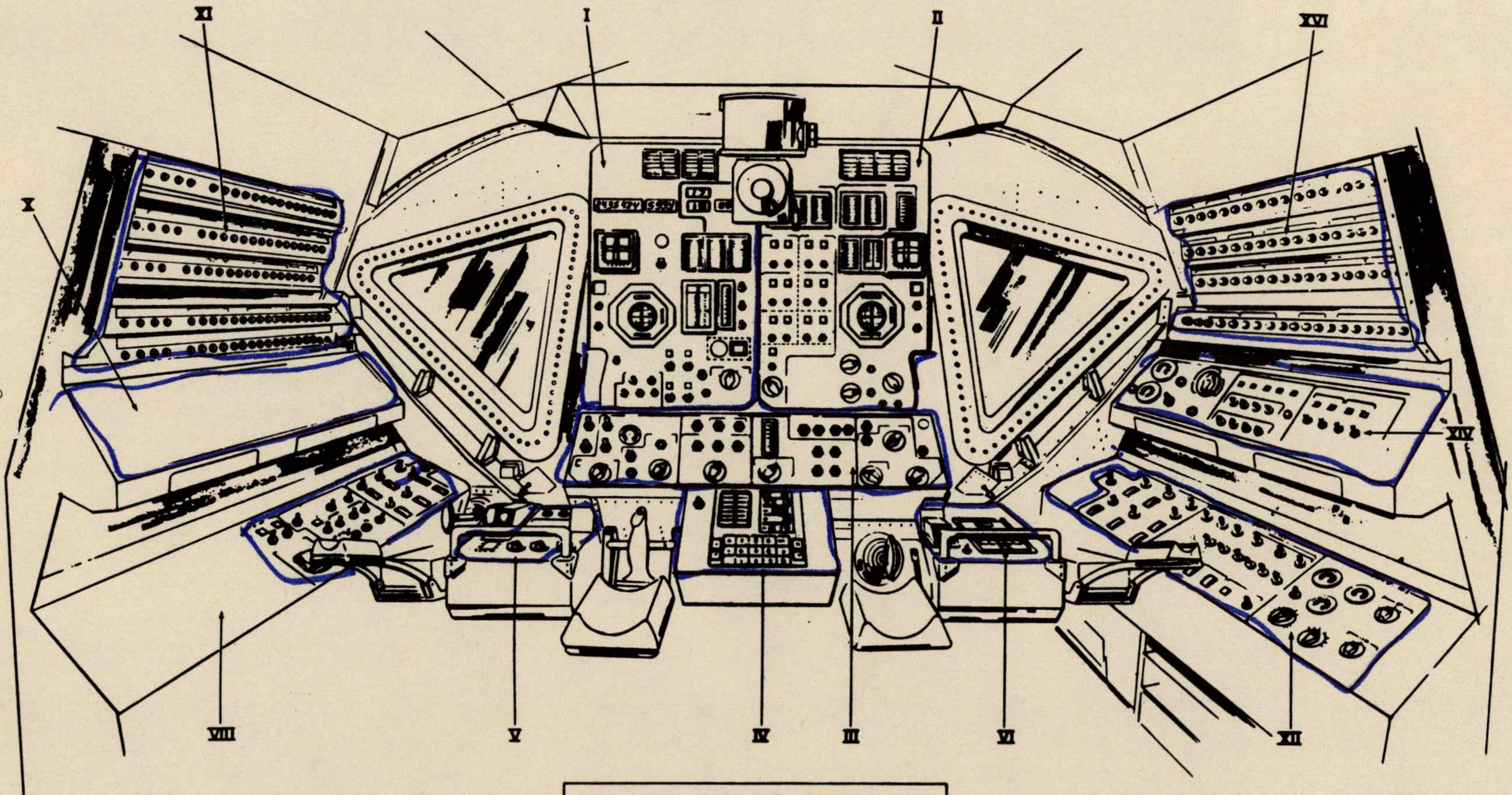
The two main display panels are constructed of two 0.015 face sheet of aluminum alloy, spaced 2 inches apart; they are reinforced by 0.015 channels. This forming a rigid box beam construction with a favorable strength-to-weight ratio and a relatively high neutral frequency. Each panel being supported to the structure by four shock mounts. Panel instruments are bezel-mounted to the back surface of the panels 2 inches along the case from the dial face, thereby achieving an attaching point approximately near the center-of-gravity of the instruments. The instrument cases are cushioned where they protrude through the panels dampening vibration tendencies. All dial faces are flush with the forward panels giving a clean display area which can be scanned readily by both astronauts. The two lower panels (3 & 4) are within easy reach and scanning for the astronauts. The two small panels (5 & 6) are in front of the astronauts at waist height. These panels contain the controls for all integral flood and signal lighting, as well as external lights, abort guidance and event timer.

At the left side of the Commander is a three-tiered console (Figure 21). The console runs fore and aft, from the front face assembly to the aft bulkhead (+Z27). The lower tier is cantered up 15° from horizontal; the center tier is cantered up 36.5° from the horizontal; the upper tier has a four angled surface containing circuit breakers, each row is cantered 15° to the line of sight. This permits a line of sight so that the white band on the circuit breakers may be readily seen when they pop.

The Systems Engineer has a three-tiered console on his right side, similar to that of the Commander's (Figure 22). All tiers are cantered at the same angle as the Commander's for ease of scanning and operation.

Compartment Deck: The crew compartment deck is approximately 36 x 55 inches, which is a seemingly small area but due to the curved construction of the compartment there is sufficient room for astronaut maneuverability. (Figures 21 and 22)

The deck construction is of aluminum honeycomb bonded to two sheets of aluminum alloy. The single construction is secured to the compartment's lower structure. Velcro-hook is bonded to deck's top



LM DISPLAYS AND CONTROLS

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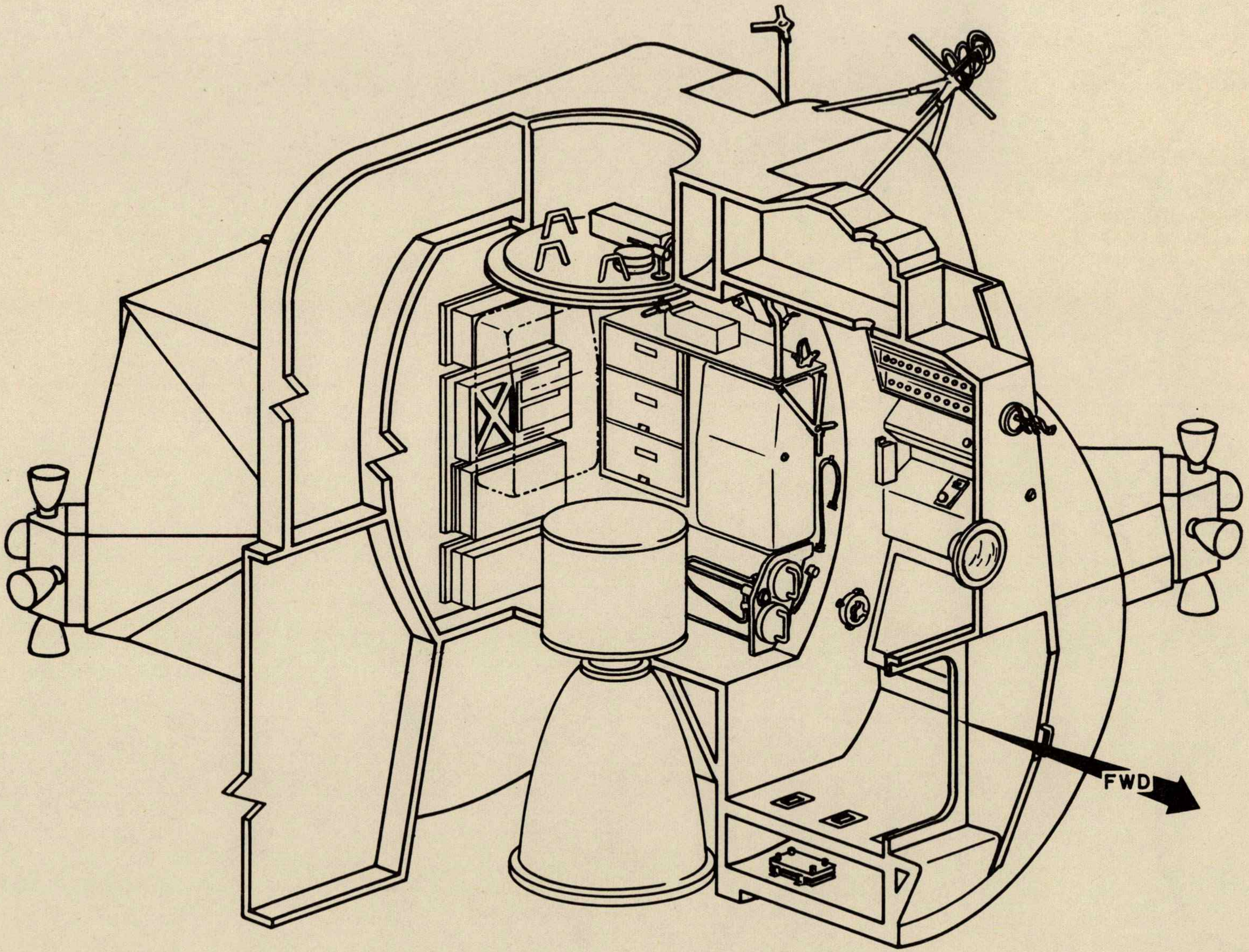
DEC 66

Figure 20

Figure 21



ASCENT STAGE-LEFT SIDE

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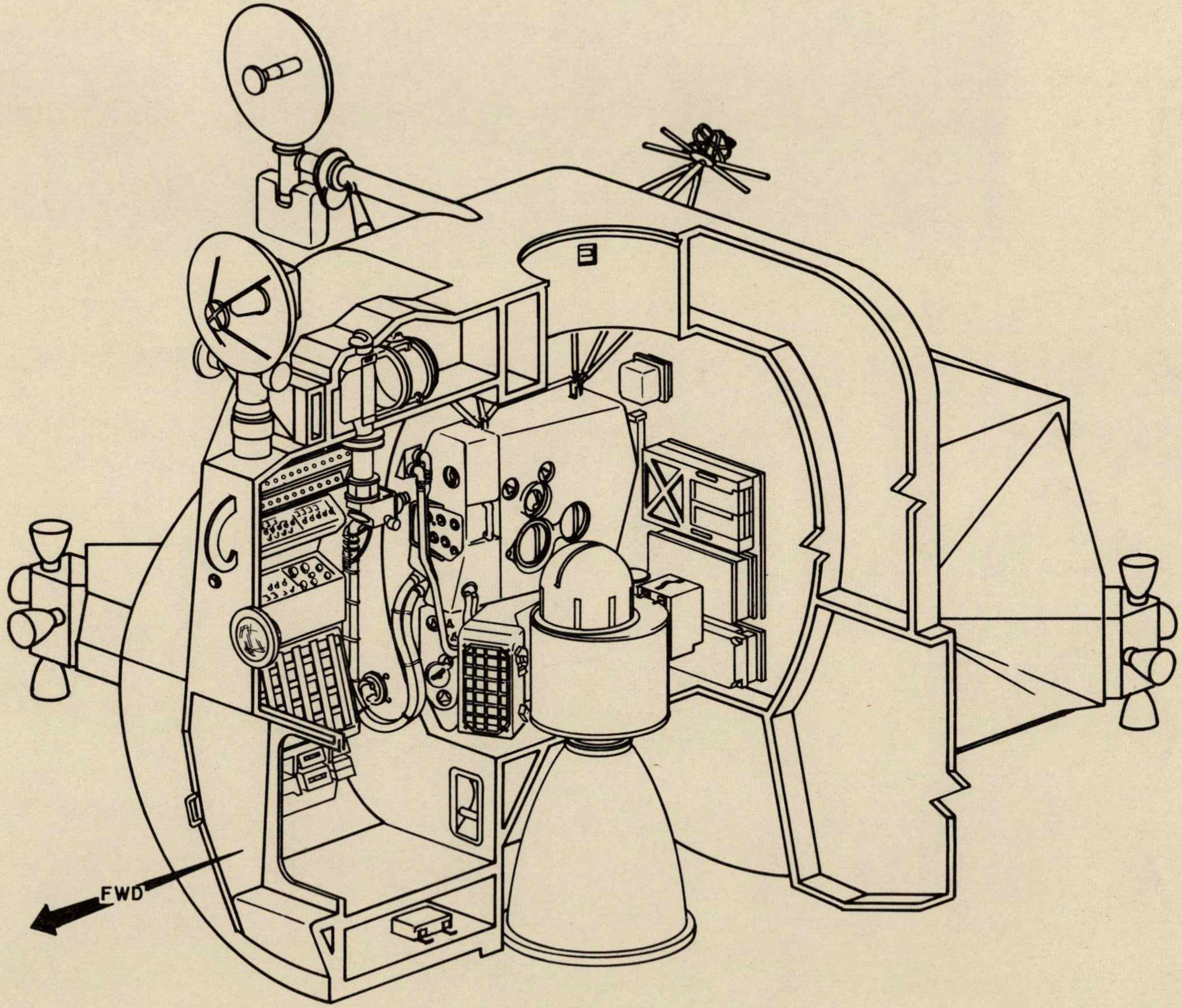

ASCENT STAGE-RIGHT SIDE

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FEB 67

Figure 22

surface, this, in conjunction with the velcro-pile on the soles of the astronaut's boots, provides a restraining force to hold the astronaut to the deck during zero "G" flight phases.

Adequate hand grips are recessed in the deck construction, to aid the astronauts, during egress and ingress maneuvers through the forward hatch.

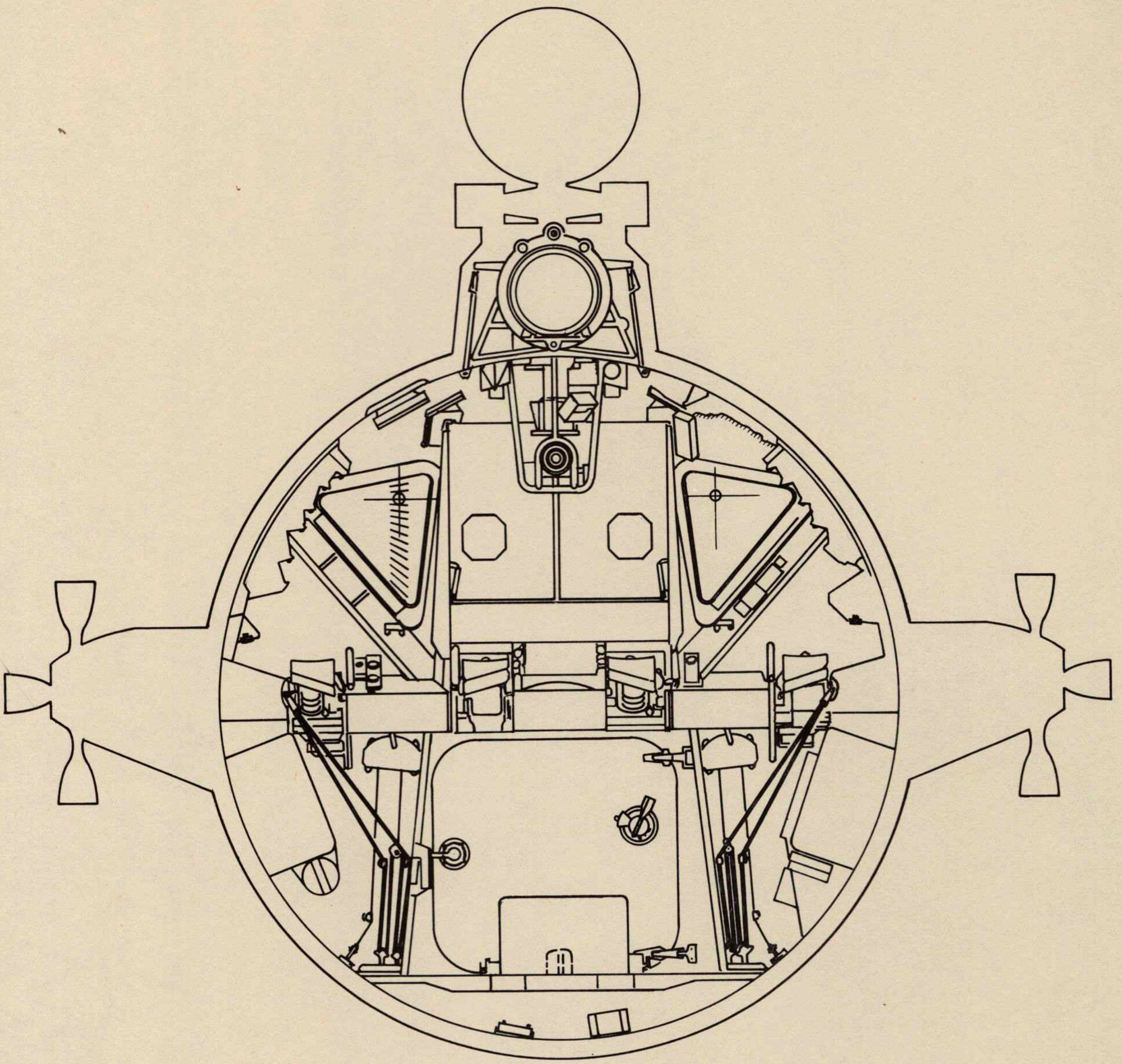
INGRESS/EGRESS HATCH

The forward hatch (Ingress/Egress) is used for transfer of the astronauts and scientific equipment to and from the LM and lunar surface. The hatch is located in the front face assembly just below the lower display panels and controls. Hand grips are provided on the forward side of the front face assembly and also recessed in the crew compartment deck, to aid the ingress and egress of the astronauts. (Figure 23)

The hatch, being approximately 32 inches square, is hinged to swing inboard when opened. A simple cam latch assembly is used to hold the hatch in the closed position; this forces a lip about the outer circumference of the hatch to enter a preloaded elastomeric silicone compound seal secured to the hatch frame. Cabin pressurization forces the hatch lip further into the seal ensuring a pressure tight contact. (Figure 24). A fixed handle is provided on the outside of the hatch for latch operation while the torque wrench (NAA Part V16-601145-11) is required to operate the latch from within the LM.

To open the hatch, it is necessary to depressurize the cabin completely. This is accomplished by opening the combination pressurization dump/relief valve within the hatch structure. A handle on the valve will open it, dumping cabin pressure and automatically shutting off cabin pressurization and turning on suit pressure. When cabin pressure reaches zero, the hatch can be opened by inserting the torque wrench and rotating the latch. The hatch can then be swung inboard. A dump valve handle is also installed on the outside of the hatch as well as the latch handle. Depressurization of the cabin and opening of the hatch can be accomplished by an astronaut, from the outside if it becomes necessary to do so.

It should be noted; in the event the torque wrench is mislaid and the hatch must be opened it can be accomplished utilizing an emergency procedure. First depressurize the cabin using the normal procedure. With pressure at zero; pull the ball-lock pin free of the latch assembly, using the cable as a puller. When free swing the latch plate assembly down, this will free the latch roller allowing the hatch



CREW COMPARTMENT
(LOOKING FORWARD)

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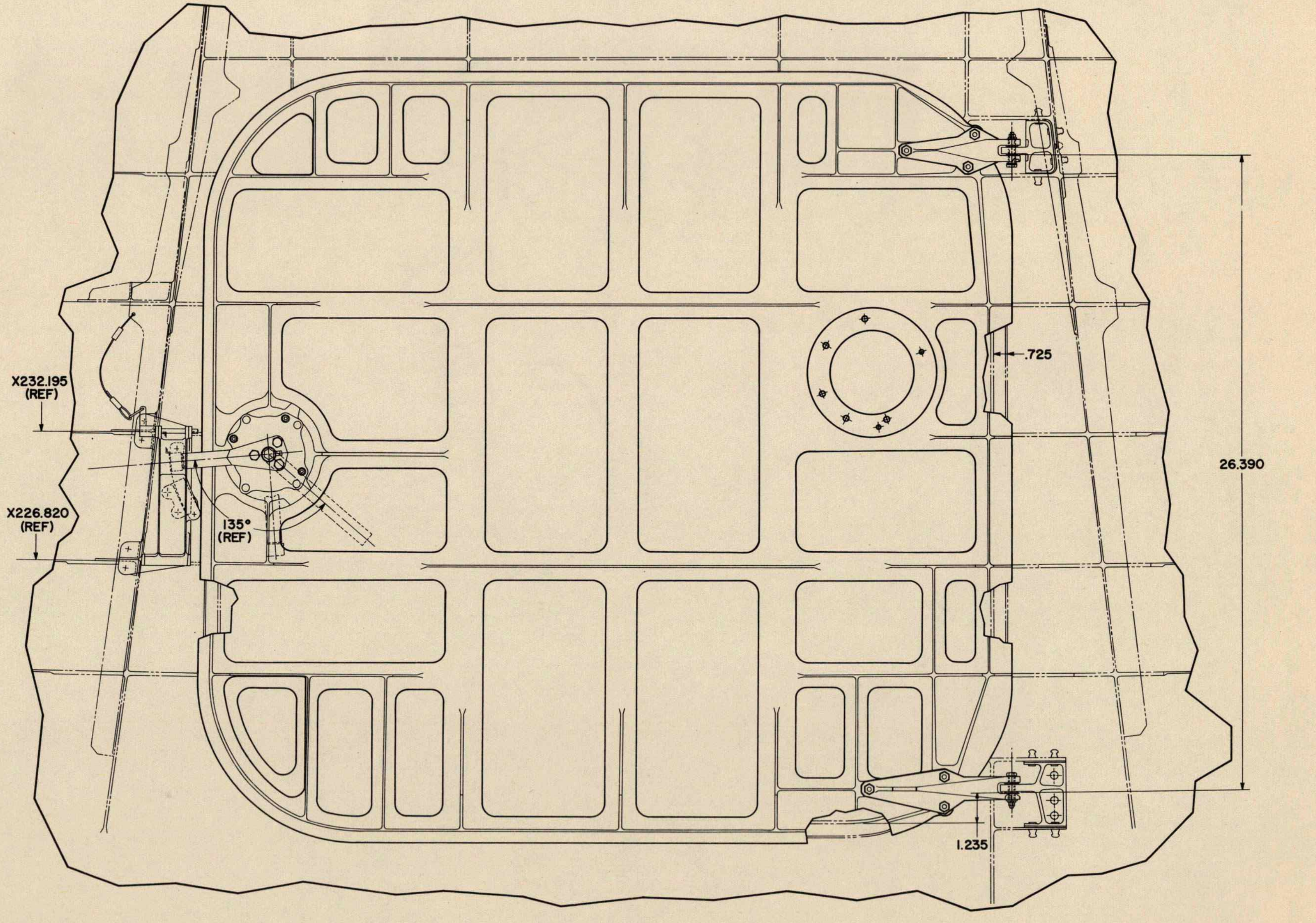
Figure 23



INGRESS/EGRESS HATCH

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APRIL 66

33

Figure 24

to be opened. To close the hatch reverse the procedure being sure the ball-lock pin is properly inserted holding the latch plate in place.

WINDOWS

Two triangular cabin windows in the front face assembly of the forward cabin section (crew compartment), provide visibility during lunar descent, lunar landing, and the rendezvous and docking phases of the mission. (Figures 23 shows location) Both windows have approximately 2 square feet of viewing area; they are cantered down to the side to permit adequate peripheral and downward visibility. (Figure 25) Each window consists of two separated panes, vented to space environment. The outer pane is made of vycor glass, coated for thermal and radiation protection; a red-blue metallic oxide being applied to the outer surface for this purpose. The inner window pane is made of a structural glass (Chemcor) designed to carry pressure loads imposed on it. The inner pane is sealed with a dual Raco seal and bolted to the structure's window frame,through an edge member bonded to the glass. The outer surface of this window is also coated with the red-blue metallic oxide. In addition, over this coating, a defogging coating is applied for thermal control. The inner surfaces are also coated; a high efficiency anti-glare metallic oxide being applied to these surfaces. (Figure 26)

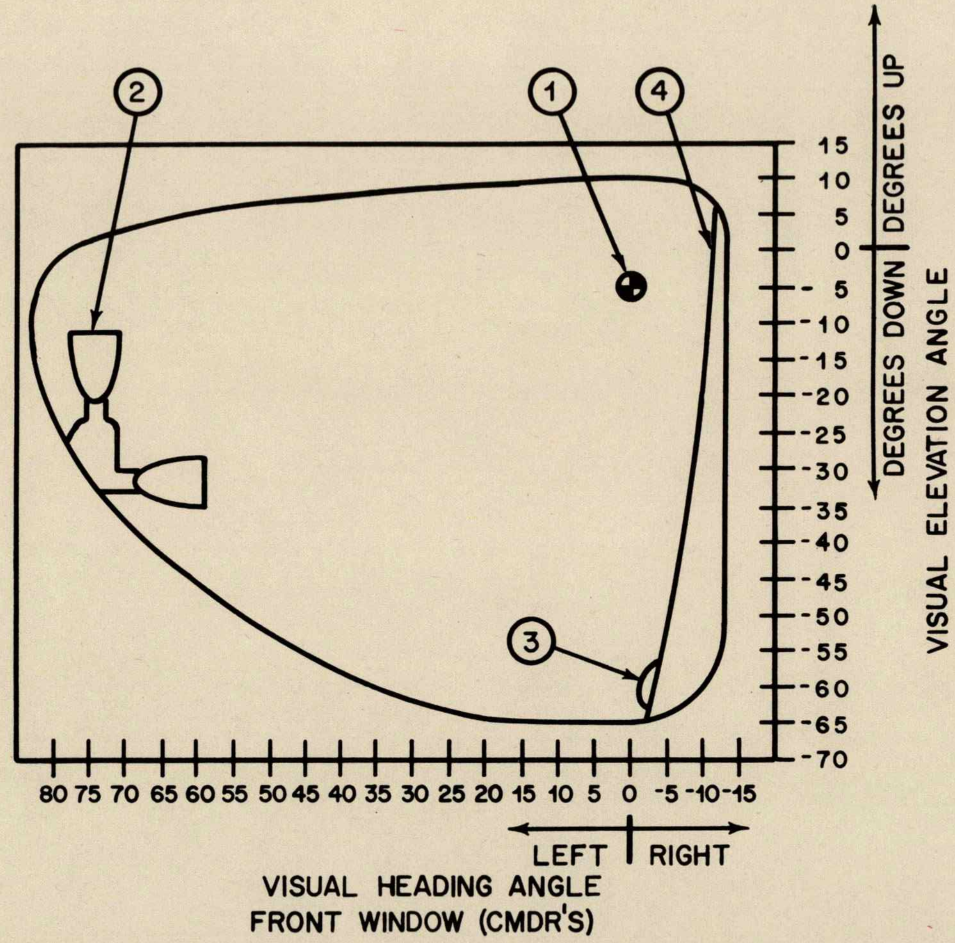
The third window is in the curved overhead bulkhead of the crew compartment, above the Commander's flight station on the left side, provides visibility for docking maneuvers. The window consists of two panes, being constructed of the same material as those of the forward. The panes are coated in the same manner with the thermal-radiation protection, anti-glare and defogging materials (Figure 27).

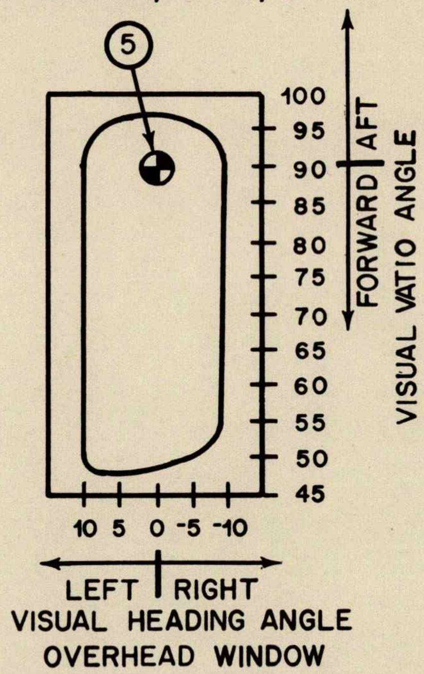
Window Coatings:

Multi-layer blue-red -- This is a metallic oxide applied to the outer surface of all window panes, by means of vacuum sputtering process at a temperature of 250°C. 36 coats are applied which in turn reduces infra-red and ultra-violet light transmission in wavelengths above and below normal visibility. (Note Wavelength Chart, Figure 28)

High Efficiency Anti-Reflection -- This is also a metallic oxide applied to the inner surface of all window panes utilizing the same process as above. The coatings reduce the mirrow effect of the windows as well as increasing their normal transmission efficiency.

1. CMDR'S DESIGN EYE
$\bar{X} = 279.25$, $\bar{Y} = 22$, $\bar{Z} = 54$
2. RCS CLUSTER
3. FRONT LANDING GEAR PAD
4. FRONT BEAM
5. CMDR'S DOCKING EYE
$\bar{X} = 280.375$, $\bar{Y} = 22$, $\bar{Z} = 37.75$





VISIBILITY DIAGRAM FROM COMMANDER'S EYE

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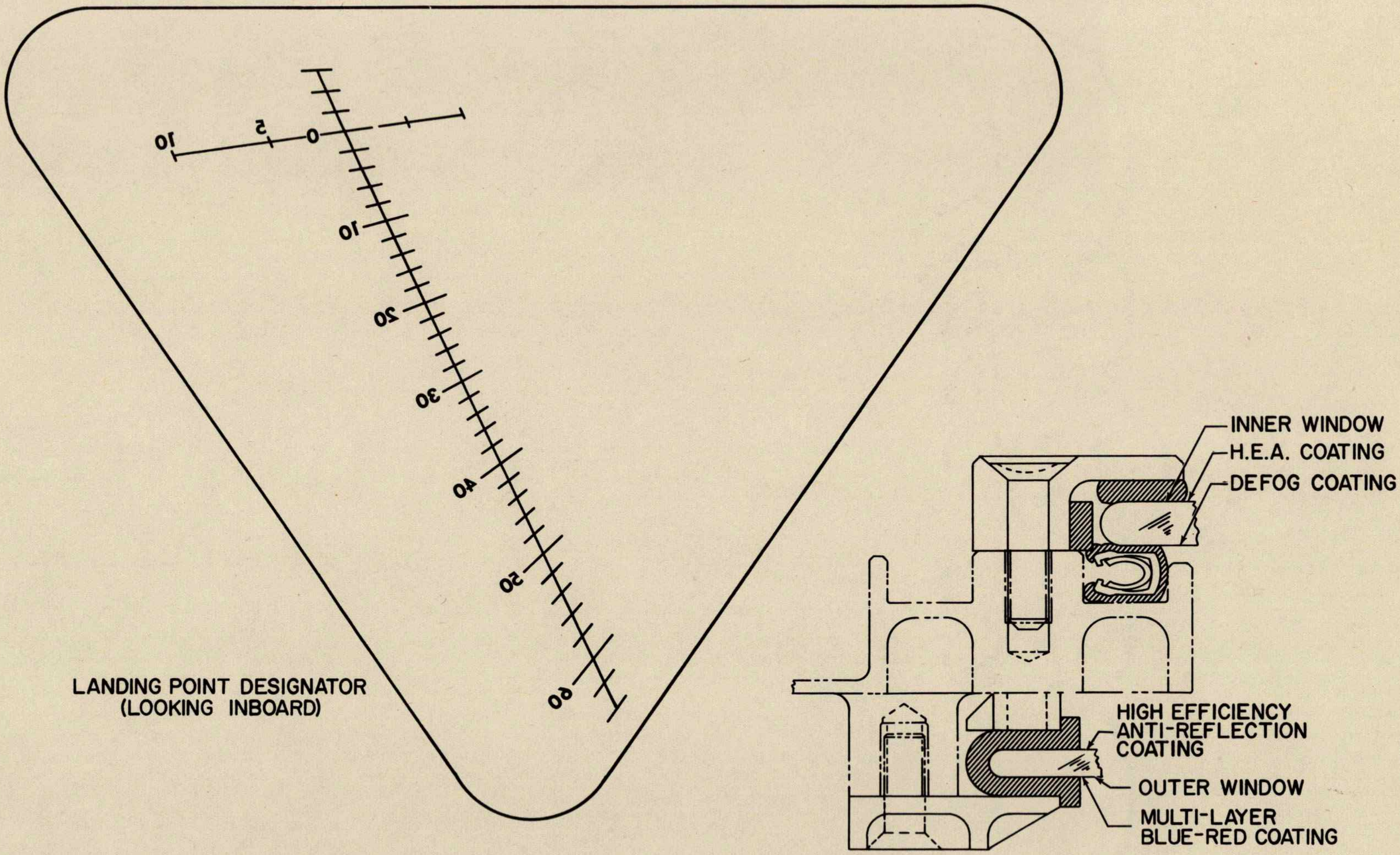
Figure 25



FRONT FACE WINDOW

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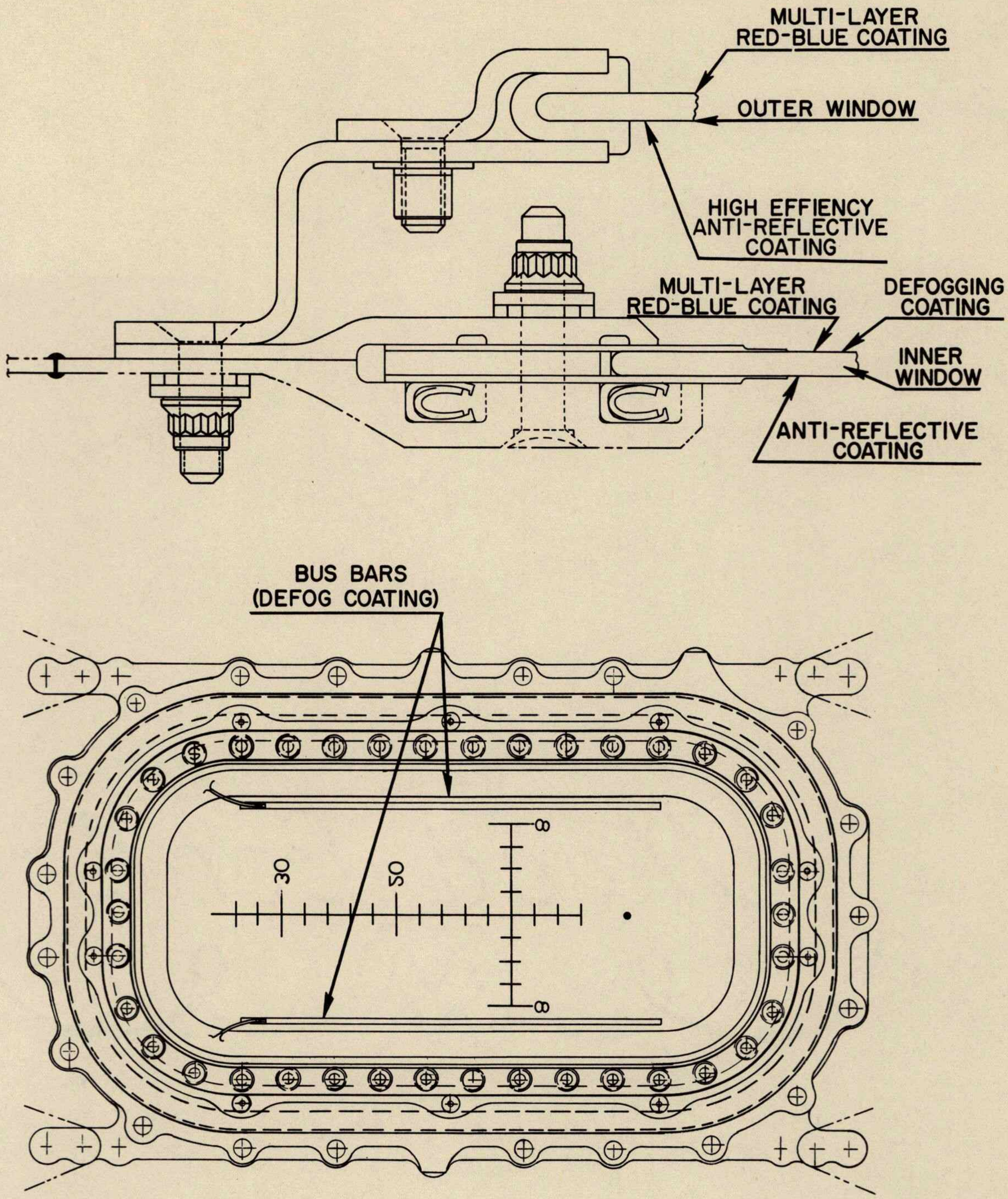
Figure 26



DOCKING WINDOW

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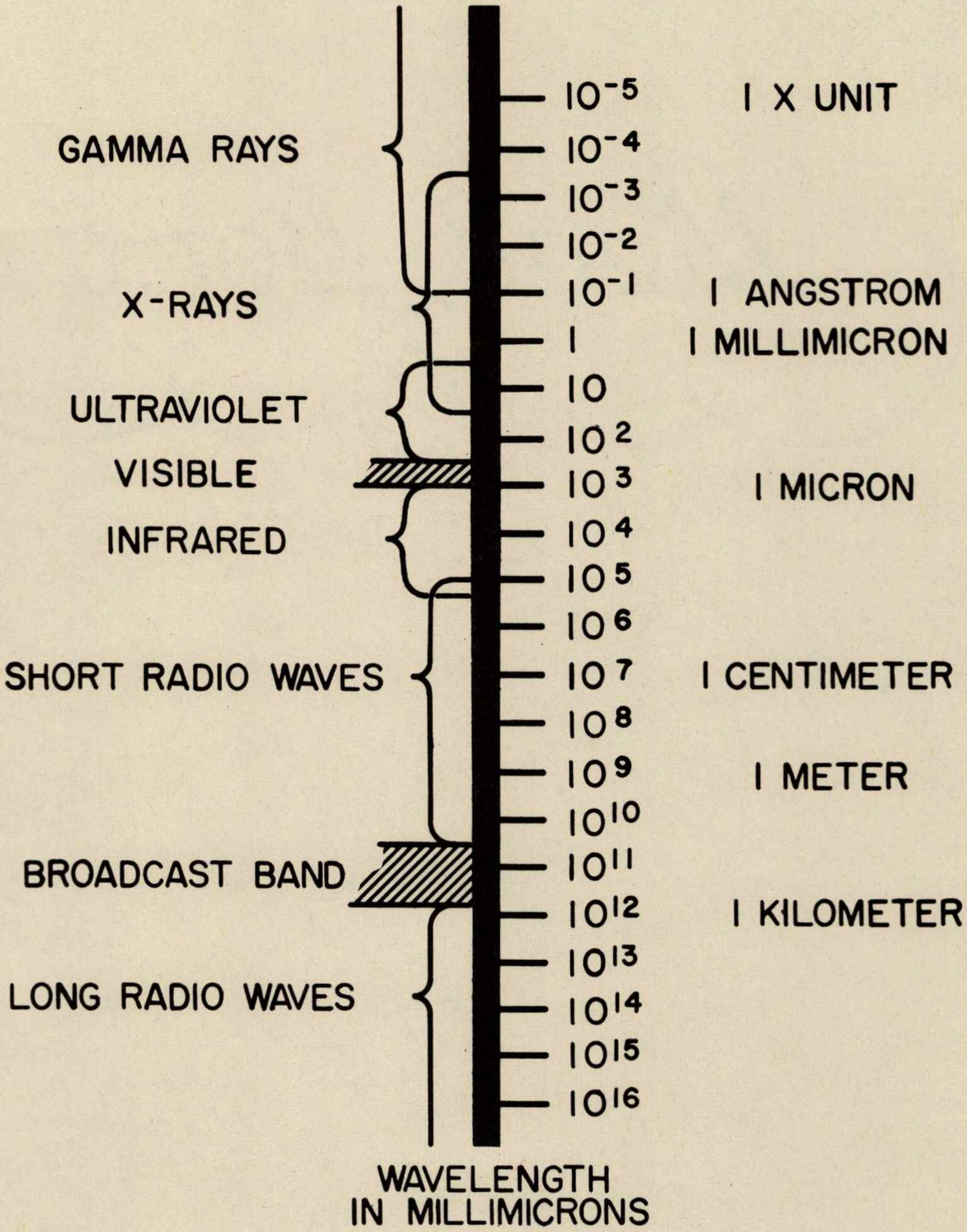
Figure 27



LIGHT TRANSMISSION IN WAVELENGTHS

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Figure 28